Thomas Puskas

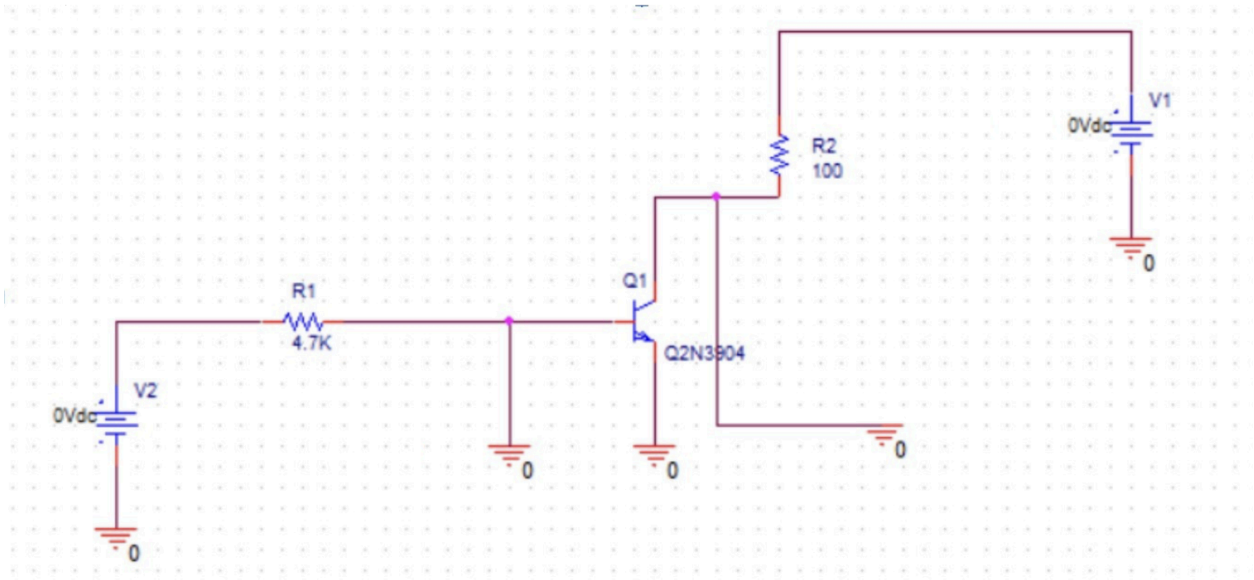
Circuits 1 Lab #05

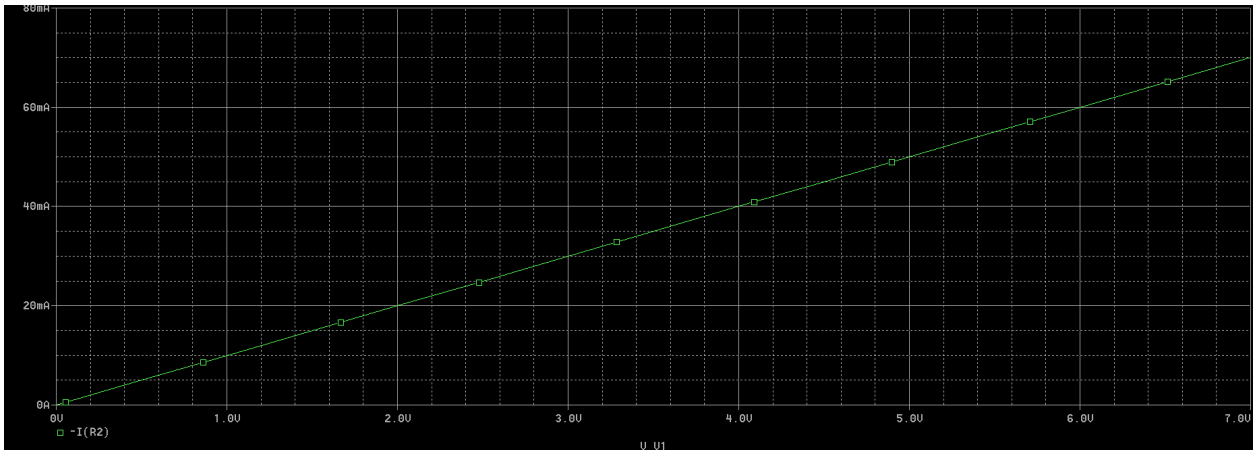
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Objective:

Through the experiment, become familiar using small and large signal analysis in a BJT circuit. This experiment also measures BJT VCE-IC characteristics in the presence of the early effect and calculates parameters of the small signal model. Then we compare the results from the small signal operation with its bias point and the small signal models.

Circuit (PSPICE):





Experiment:

3.1 From the equation, calculate the parameters

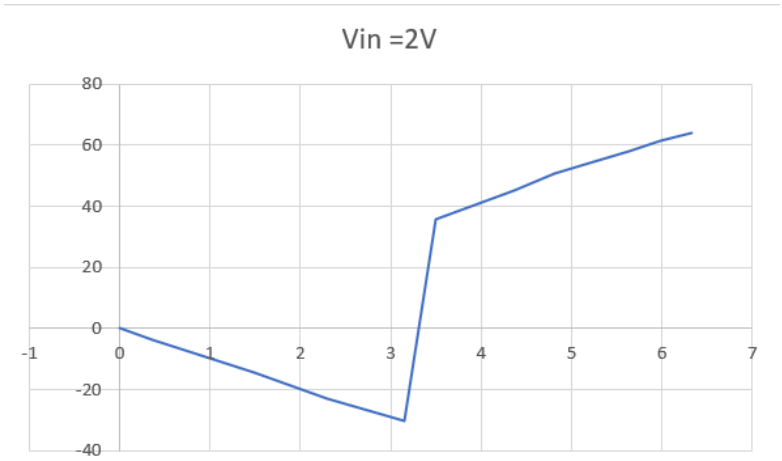
Procedure:

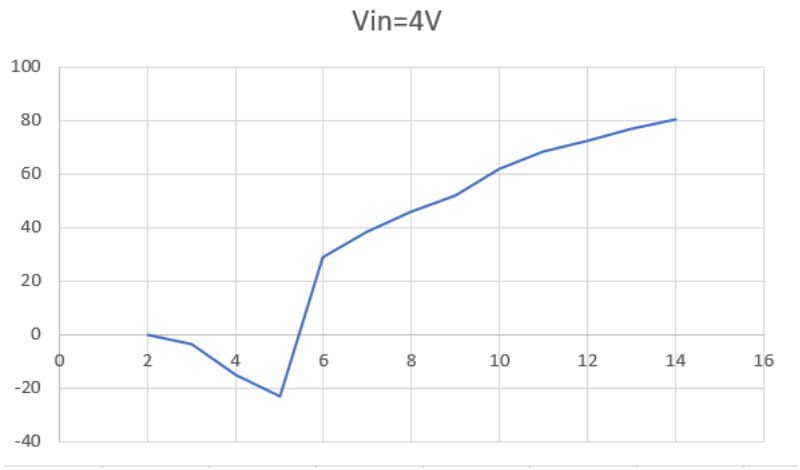
1. Connect the circuit with RB =4.7kΩ, RC= 100Ω, Q1 =2N3904.
2. Set VCC to 0V.
3. Set M1 and M2 to the highest milliampere range to make sure the meters are protected.
4. After the power is applied, the range is selected
5. Then connect a signal generator to Vin input termnials
6. Then set output of signal generator to 2V DC and close S1
7. Vcc then varies from 0V to 12V and then observe and record the collector voltage VCE and the collector current IC and the base current IB

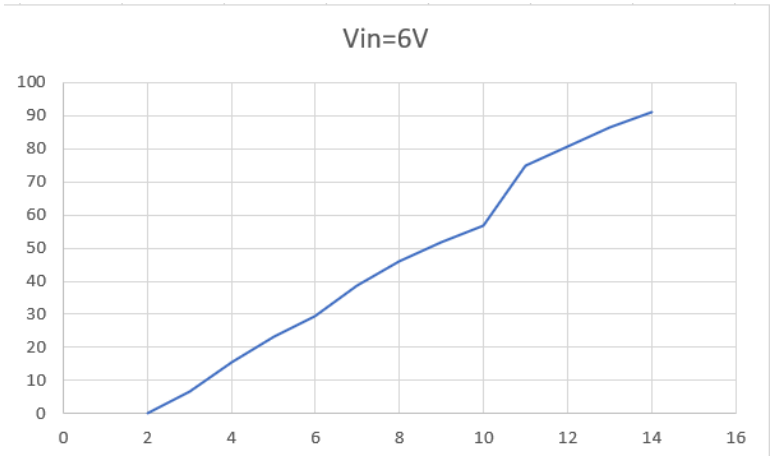
Data Table:

V_{CC}[V]	Step 2: $V_{in} = V_{IN}$=2V			Step 4: $V_{in} = V_{IN}$=4V			Step 5: $V_{in} = V_{IN}$=6V		
	V_{CE}	I_B	I_C	V_{CE}	I_B	I_C	V_{CE}	I_B	I_C
0	-0.86m	279.8µ	0.65µ	-0.13m	0.693m	0.39µ	-0.28m	1.11m	-2.67µ
1	354.6m	272.2µ	-3.71m	0.75V	0.689m	-3.79µ	415.3m	1.10m	6.6m
2	1.49V	260.0µ	-14.4m	1.51V	0.682m	-15.33µ	1.47	1.10m	15.4m
3	2.29V	257.8µ	-22.8m	2.38V	0.67m	-23.18µ	2.26	1.098m	23.2m
4	3.15V	257.4µ	-30.3m	3.20V	0.677m	29.0µ	3.01	1.045m	29.60m
5	3.50V	257.0µ	35.6m	4.04V	0.674m	38.57µ	3.85	1.05m	38.54m
6	3.93V	257.4µ	40.45m	4.87V	0.672m	45.84µ	4.75	1.08m	46.15m
7	4.36V	258.9µ	45.18m	5.66V	0.674m	51.73µ	5.53	1.09m	51.73m
8	4.81V	260.3µ	50.75m	6.28V	0.670m	62.05µ	6.36	1.09m	56.60m
9	5.24V	262.5µ	54.63m	6.74V	0.677m	68.18µ	7.19	1.09m	74.74m
10	5.64V	264.4µ	58.19m	7.15V	0.680m	72.37µ	7.83	1.08m	80.80m
11	5.99V	268.4µ	61.54m	7.55V	0.682m	76.76µ	8.41	1.08m	86.31m
12	6.33V	277.8µ	64.24m	7.95V	0.682m	80.49µ	8.86	1.09m	90.91m

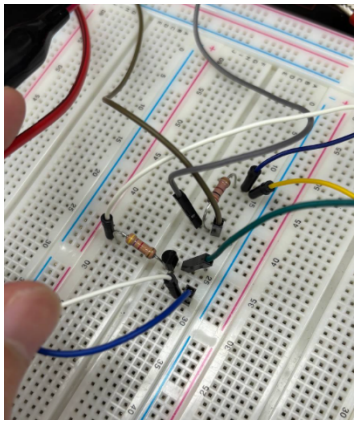
Plotting of graph


Vin =2V
80
60
40
20
0
-20
-40
-1
0
1
2
3
4
5
6
7


Vin=4V
100
80
60
40
20
0
-20
-40
0
2
4
6
8
10
12
14
16

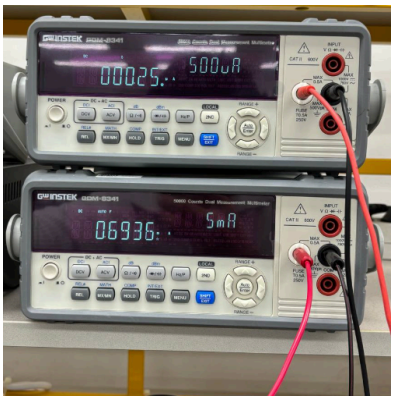

Vin=6V
100
90
80
70
60
50
40
30
20
10
0
0
2
4
6
8
10
12
14
16

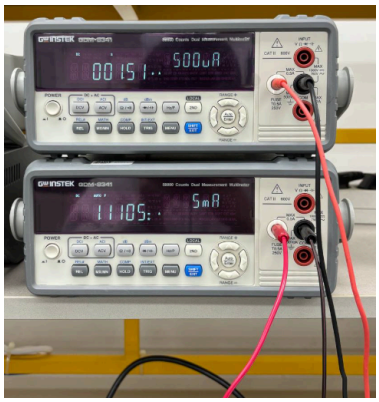
Circuit:



Obtaining data.











Transconductance: gm=2.47.

Small Signal Resistance: 38.12

The early voltage is 75V and the small-signal output resistance is 1.85kΩ, and the gain of small-signal transfer function is 150V.

3.2 Measuring the small-signal gain of BJT circuit.

Procedure:

1. Adjust the generator output to 1kHz sine wave signal for 2V DC voltage and a minimum amplitude of AC voltage.
2. Measured and record the peak-to-peak voltage of vin and vout.
3. Set the DC voltage of the signal generator to 4V and repeat the steps.
4. Do for 6V.

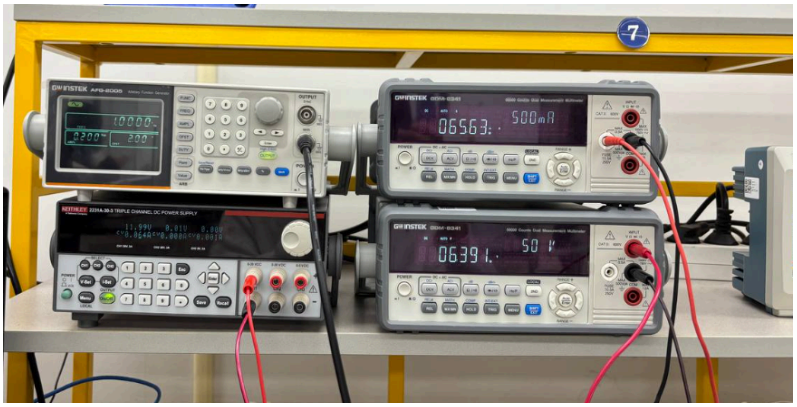
Data:

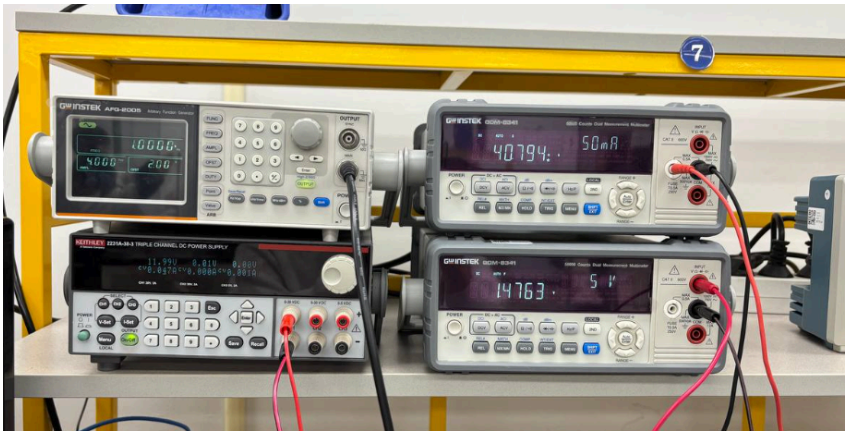
	Vin	vout	Av
Vin =2V DC	1.354	6.52	4.815
Vin =4V DC	3.347	8.01	2.393
Vin =6V DC	5.33	9.00	1.688

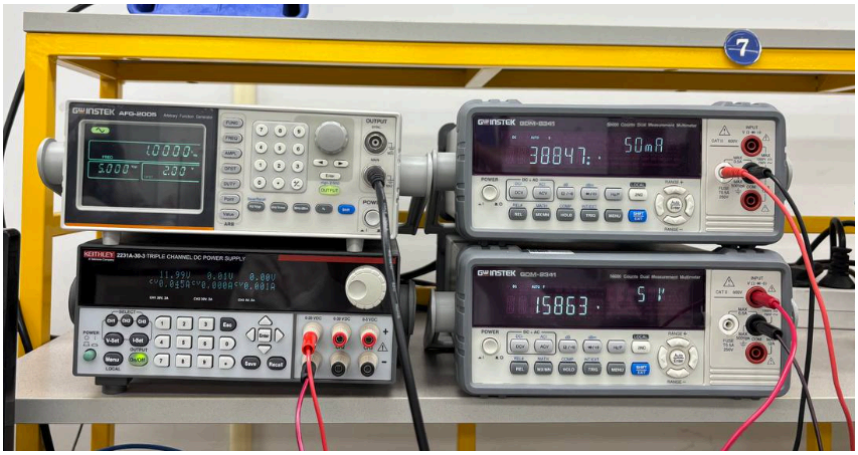
After, turn the DC voltage of the signal generator to 2V and adjust the AC voltage 100m, 200m, 500m, 1V, 2V, 3V, 4V, 5V.

Data

AC voltage Vin	Vin	Vout	Av
0.1	1.362	6.38	4.684
0.2	1.356	6.38	4.705
0.5	1.353	6.38	4.715
1	1.342	6.32	4.709
2	1.382	5.98	4.327
3	1.432	5.21	3.638
4	1.476	4.80	3.252
5	1.587	4.66	2.936








7
GW INSTEK
AFG-2005
OUTPUT
GDM-8341
50mA
KEITHLEY
2231A-30-3 TRIPLE CHANNEL DC POWER SUPPLY